Example 5.3

Calculate the regression equation of X on Y from the data given below, taking deviations from actual means of X and Y.

x	12	14	15	14	18	17
y	42	40	45	47	39	45

Estimate the likely demand when the $X = 25$.

Solution:

	x_i	$u_i = x_i - 15$	u_i^2	y_i	$v_i = y_i - 43$	v_i^2	$u_i v_i$
	12	-3	9	42	-1	1	3
	14	-1	1	40	-3	9	3
	15	-0	0	45	2	4	0
	14	-1	1	47	4	16	-4
	18	3	9	39	-4	16	-12
	17	2	4	45	2	4	4
Total	90	0	24	258	0	50	-6

$$\bar{x} = \sum_{i=1}^{6} x_i / 6 = \frac{90}{6} = 15$$

$$\bar{y} = \sum_{i=1}^{6} y_i / 5 = \frac{258}{6} = 43$$

The regression line of U on V is computed as under

$$\hat{b}_{UV} = \frac{n\sum_{i=1}^{n} u_i v_i - \sum_{i=1}^{n} u_i \sum_{i=1}^{n} v_i}{n\sum_{i=1}^{n} v_i^2 - \left(\sum_{i=1}^{n} v_i\right)^2} = \frac{6(-6)}{6 \times 50} = -0.12$$

$$\hat{a} = \bar{u} - \hat{b}_{UV}\,\bar{v} = 0$$

Hence, the regression line of U on V is $U = \hat{b}_{UV}\, v + \hat{a} = -0.12v$

Thus, the regression line of X on Y is $(Y-43) = -0.25(x-15)$

When $x = 25$, $y - 43 = -0.25\ (25-15)$

$$y = 40.5$$

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